LPC2148 – ADC & DAC



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ADC

- Two , 10-bit successive approximation ADCs

ADC0 & ADC1
ADC0 = 6 Channels
ADC1 = 8 Channels
Total = 14 channels

- Measurement range 0 V to VREF (3.3V)
- 10-bit conversion time ≥ 2.44 µs
- Burst conversion mode for single or multiple inputs.
- Optional conversion on transition on input pin or Timer Match signal
- Burst conversion mode for single or multiple inputs

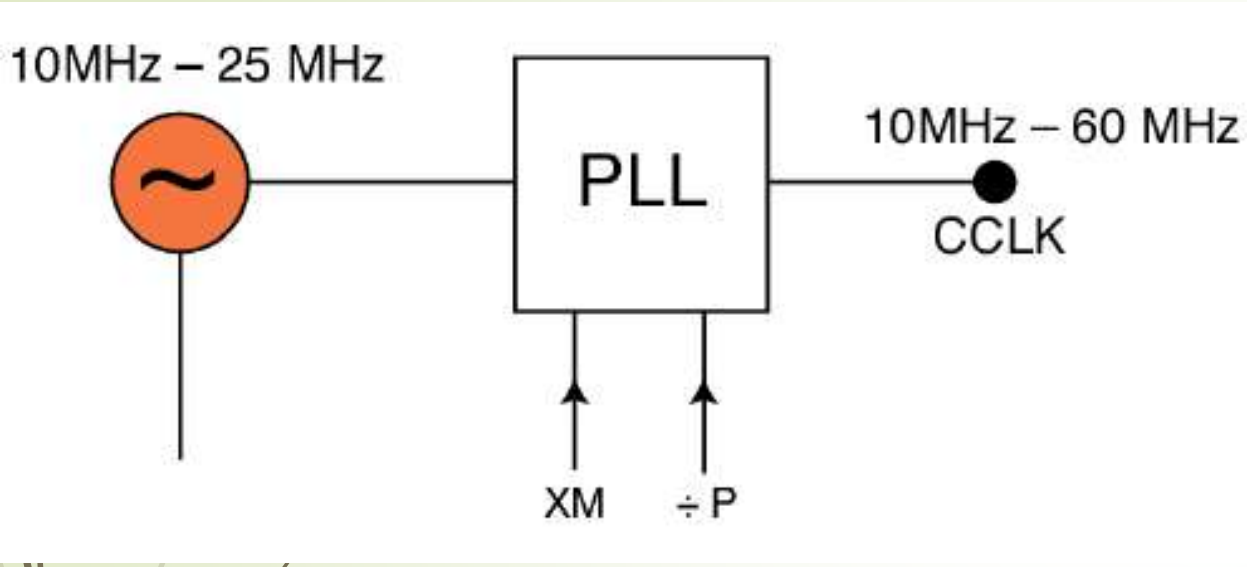
DAC

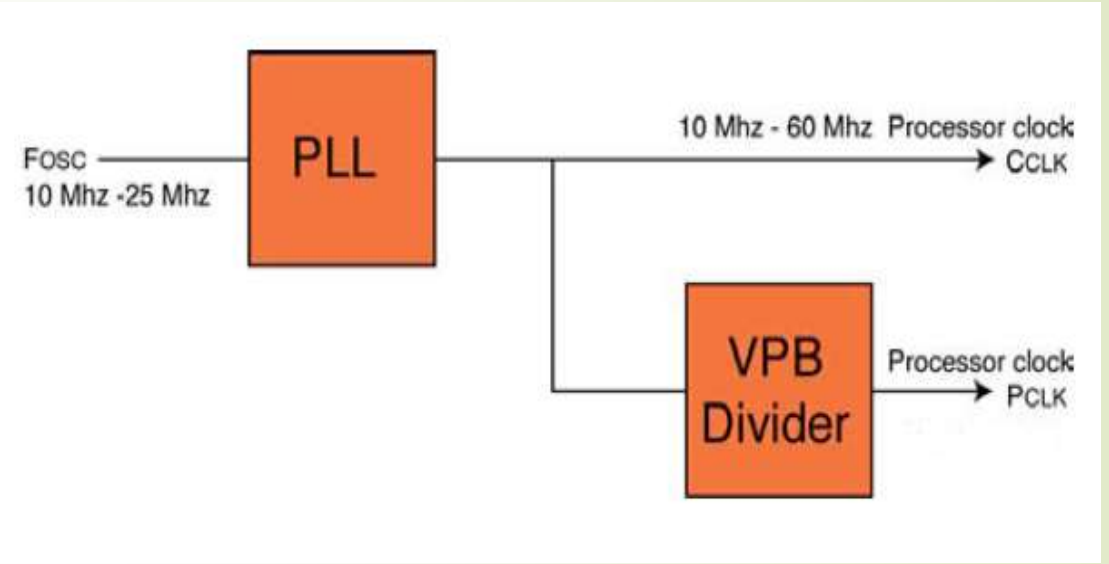
- **10-bit DAC**

The DAC enables to generate a variable analog output. The maximum DAC output voltage is the VREF voltage.

- Buffered output.
- Power-down mode available.
- Selectable speed versus power.

PLL





VPBDIV = 0x00000001;

Bit	Symbol	Value	Description	Reset value
1:0	VPBDIV	00	VPB bus clock is one fourth of the processor clock.	00
		01	VPB bus clock is the same as the processor clock.	
		10	VPB bus clock is one half of the processor clock.	
		11	Reserved. If this value is written to the VPBDIV register, it has no effect (the previous setting is retained).	

Pins Involved

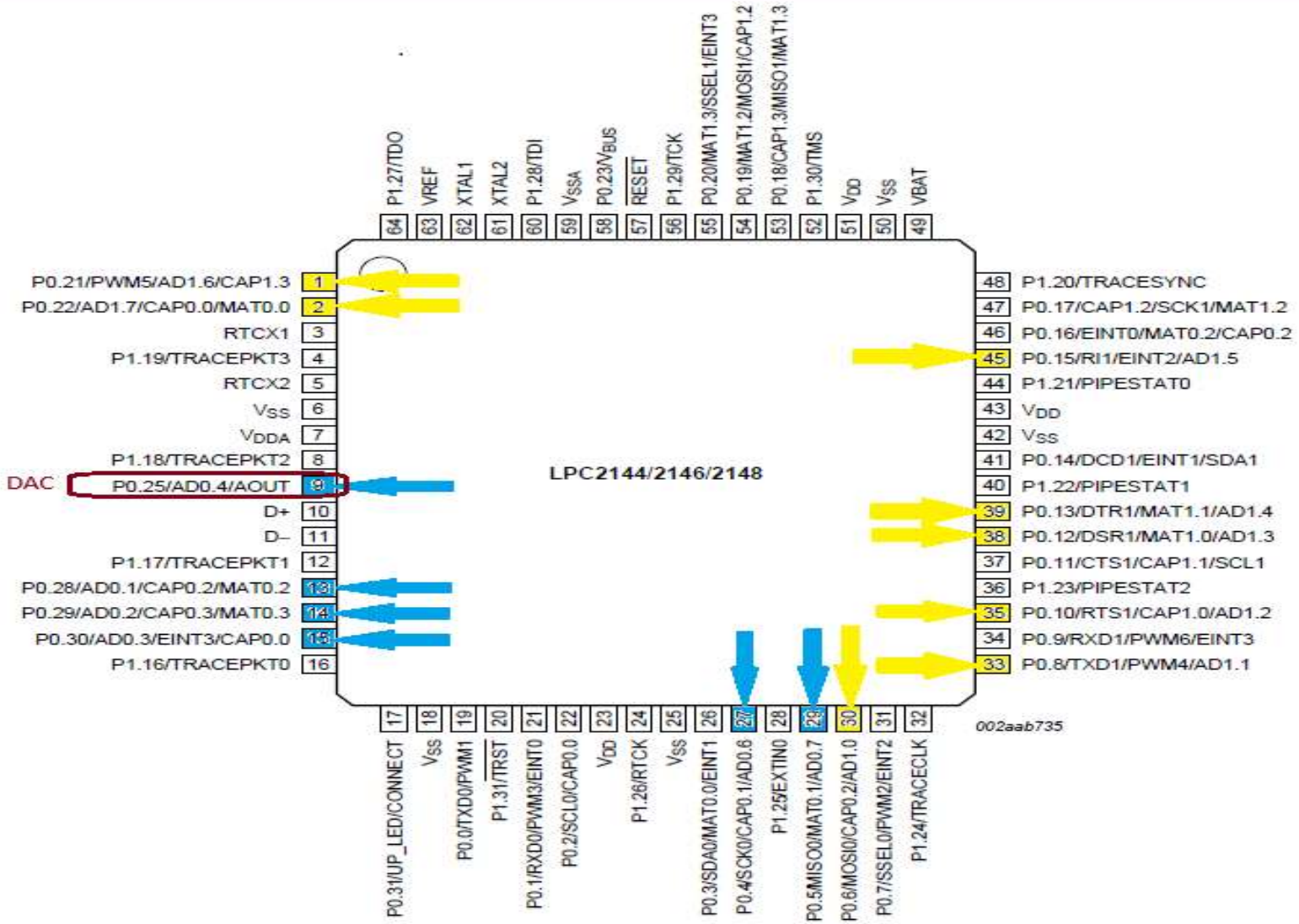


Fig 4. LPC2144/46/48 pinning

ADC0

ADC0.1 [Pin 13] P0.28

ADC0.2 [Pin 14] P0.29

ADC0.3 [Pin 15] P0.30

ADC0.4 [Pin 9] P0.25 - [DAC Pin]

ADC0.6 [Pin 27] P0.4

ADC0.7 [Pin 29] P0.5

ADC1

ADC1.0 [Pin 30] P0.6

ADC1.1 [Pin 33] P0.8

ADC1.2 [Pin 35] P0.10

ADC1.3 [Pin 38] P0.12

ADC1.4 [Pin 39] P0.13

ADC1.5 [Pin 45] P0.15

ADC1.6 [Pin 1] P0.21

ADC1.7 [Pin 2] P0.22

SFRs Involved

- VLSI Peripheral Bus Divider - VPBDIV
- P0 Pin Select Register - PINSEL1
- ADC Control Register - ADCR
- ADC Data Register - ADDR
- DAC Control Register - DACR

Pin Select SFRs

- PINSEL0 for GPIO pins 0.0 to 0.15
- PINSEL1 for GPIO pins 0.16 to 0.31
- PINSEL2 for GPIO pins 1.16 to 1.31

Each Pin is Associated with multiple functions

Pin Select enable you to select which pin functions you would like to use

How to Use PINSEL

- Each associated 'pin' in PINSEL is assigned a 2-bit address.

PINSEL1

Pin	0.31	0.30	0.29	0.28	0.27	0.26	0.25	0.24	0.23	0.22	0.21	0.20	0.19	0.18	0.17	0.16
Bits	31..30	29..28	27..26	25..24	23..22	21..20	19..18	17..16	15..14	13..12	11..10	9..8	7..6	5..4	3..2	1..0

```
PINSEL1 = PINSEL1|0x00001000;
```

```
PINSEL1 = PINSEL1 | 2 << (25-16) * 2
```

ADC Control Register - ADCR

Function	-	EDGE	START	-	PDN	-	CLKS	BURST	CLKDIV	SEL
ADCR Bit(s)	31..28	27	26..24	23.22	21	20	19..17	16	15..8	7..0

SEL

1	0	0	0	0	0	0	0



Channel - 7

- This set of 8 bits corresponds to the 8 different 'channels' available on either A/D converter.
- You can indicate which channel you wish to use by setting it's appropriate bit to '1‘

CLKDIV

- **The A/D Converters on the LPC2148 are able to run at a maximum speed of 4.5MHz**.
- The conversion speed is selectable by the user
- To arrive at a number equal to or less than 4.5MHz, we need to 'divide' our PCLK (the speed at which our microprocessor is running) by a fixed number, which we provide (in binary format) using the 8 **CLKDIV** bits.
- **In order to avoid a 'divide by 0' error, though, the A/D control register will add one to whatever value you supply.**
- **Pclk = 60 MHz, Div = 60/4.5 = 14 ➔ 14-1 = 13**

0	0	0	0	1	1	0	1

Burst

- Set this to 1 for doing repeated conversions.
- Set this bit to 0 for software-controlled conversions , which take 11 clocks to finish. Here is a remark from datasheet : START bits must be set to 000 when BURST=1 or conversions will not start. Refer datasheet for detailed info.

	-	EDGE	START	-	PDN	-	CLKS	BURST	CLKDIV	SEL
AD0_CR	****	*	***	**	*	*	000	*	********	********

0	0	0



10 Bit ADC

CLKS

- Used to Set Resolution of ADC

```
000 = 10-bits (0..1023)
001 =  9-bits (0..511)
010 =  8-bits (0..255)
011 =  7-bits (0..127)
100 =  6-bits (0..63)
101 =  5-bits (0..31)
110 =  4-bits (0..15)
111 =  3-bits (0..7)
```

	-	EDGE	START	-	PDN	-	CLKS	BURST	CLKDIV	SEL
AD0_CR	****	*	***	**	*	*	000	*	********	********

0	0	0



10 Bit ADC

PDN ➔ ADC ON/OFF

- Used to Set whether the ADC is in Power Down Mode (Off) or Involved in data conversion (ON)

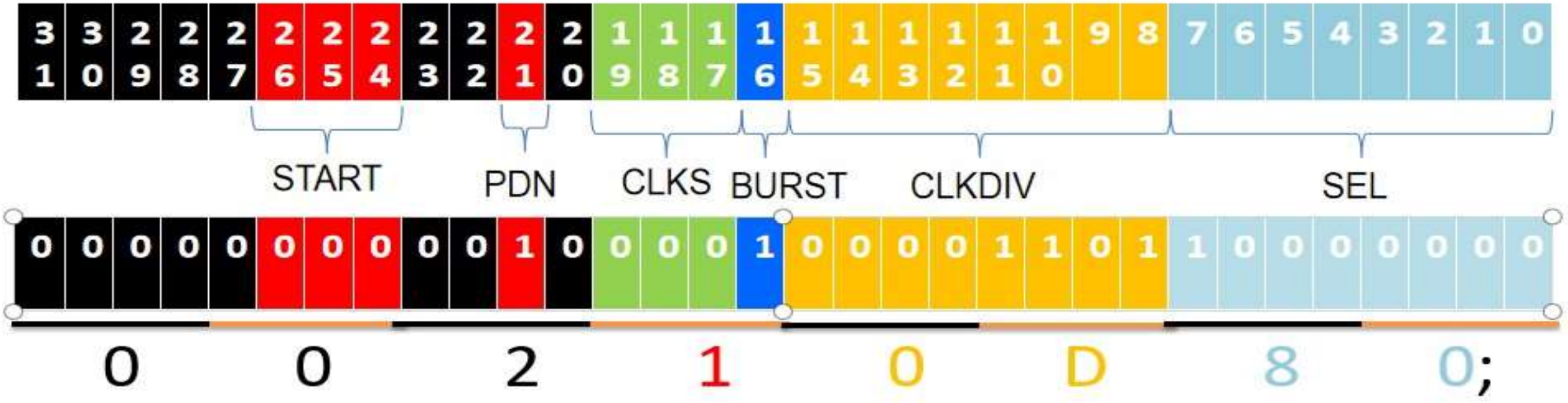
PDN

PDN (short for 'Power-Down') indicates whether the ADC should be in 'Power-Down' mode (0), or actively converting data (1). Since we want to convert data right away, we could tell the ADC to go out of power-down mode (it's default value) by setting this bit to 1 as follows:

	-	EDGE	START	-	PDN	-	CLKS	BURST	CLKDIV	SEL
AD0_CR	****	*	***	**	1	*	***	*	********	********

1 → **Turn ON ADC**

ADC1 Control Register



	-	EDGE	START	-	PDN	-	CLKS	BURST	CLKDIV	SEL
AD0_CR	****	*	***	**	1	*	***	*	********	********

Starting the Conversion

Table 255: A/D Control Register (AD0CR - address 0xE003 4000 and AD1CR - address 0xE006 0000) bit description

Bit	Symbol	Value	Description	Reset value
26:24	START		When the BURST bit is 0, these bits control whether and when an A/D conversion is started:	0
		000	No start (this value should be used when clearing PDN to 0).	
		001	Start conversion now.	
		010	Start conversion when the edge selected by bit 27 occurs on P0.16/EINT0/MAT0.2/CAP0.2 pin.	
		011	Start conversion when the edge selected by bit 27 occurs on P0.22/TD3/CAP0.0/MAT0.0 pin.	
		100	Start conversion when the edge selected by bit 27 occurs on MAT0.1.	
		101	Start conversion when the edge selected by bit 27 occurs on MAT0.3.	
		110	Start conversion when the edge selected by bit 27 occurs on MAT1.0.	
		111	Start conversion when the edge selected by bit 27 occurs on MAT1.1.	

Bits (26:24) = 001

AD1GDR – ADC1 Global Data Register

17.4.2 A/D Global Data Register (AD0GDR - 0xE003 4004 and AD1GDR - 0xE006 0004)

Table 256: A/D Global Data Register (AD0GDR - address 0xE003 4004 and AD1GDR - address 0xE006 0004) bit description

Bit	Symbol	Description	Reset value
5:0	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
15:6	RESULT	When DONE is 1, this field contains a binary fraction representing the voltage on the Ain pin selected by the SEL field, divided by the voltage on the V_{DDA} pin (V/V_{REF}). Zero in the field indicates that the voltage on the Ain pin was less than, equal to, or close to that on V_{SSA}, while 0x3FF indicates that the voltage on Ain was close to, equal to, or greater than that on V_{REF}.	NA
23:16	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
26:24	CHN	These bits contain the channel from which the RESULT bits were converted (e.g. 000 identifies channel 0, 001 channel 1...).	NA
29:27	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
30	OVERUN	This bit is 1 in burst mode if the results of one or more conversions was (were) lost and overwritten before the conversion that produced the result in the RESULT bits. This bit is cleared by reading this register.	0
31	DONE	This bit is set to 1 when an A/D conversion completes. It is cleared when this register is read and when the ADCR is written. If the ADCR is written while a conversion is still in progress, this bit is set and a new conversion is started.	0

DAC Control Register - DACR

Bit	Symbol	Value	Description	Reset value
5:0	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
15:6	VALUE		After the selected settling time after this field is written with a new VALUE, the voltage on the A_{OUT} pin (with respect to V_{SSA}) is VALUE/1024 * V_{REF}.	0
16	BIAS	0	The settling time of the DAC is 1 μs max, and the maximum current is 700 υA.	0
		1	The settling time of the DAC is 2.5 μs and the maximum current is 350 μA.	
31:17	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA